



BrightLearn Data Analytics Research Assignment 1

Foundations of Data Analytics & Data Science

Course	Data Analytics & Data Science
Assignment No.	Research Assignment 1
Target Audience	Beginner Students
Due Date	To be confirmed by instructor
Format	Handwritten answers — scan and email your completed work

Instructions

Read each question carefully and research the answer before writing.

- All answers must be handwritten — typed submissions will not be accepted.
- Write in full sentences and use your own words.
- Include diagrams or examples where relevant — this will earn you bonus marks.
- Cite your sources at the bottom of each section.
- Scan your completed assignment and submit it before the due date.

Section A: Data Fundamentals



Research Tip: Start with official documentation from platforms like IBM, Google Cloud, AWS, or Microsoft Azure — they offer excellent beginner-friendly explanations.

1. What is a Database? Define it in your own words and explain why databases are essential in the modern world. Provide two real-world examples of where databases are used.

A database is a place where data is stored. It is a top level container wherein everything for a company is



System lives in.

- Data/Information is currency in the modern world, so it's important to store a secure data in a central place.
- Discovery is a great example of the use of data for innovation. Another is all insurance companies going into ~~banking~~ ^{banking using customer data}.

2. What is a Data Warehouse? How does it differ from a regular database? Why would a company use a data warehouse?

- A data warehouse is a central system used to collect and store large amounts of data from different sources.
- Databases are designed to support day-to-day transactions while data warehouses are for analysing historical data.
- Companies use data warehouses to combine info from different systems (databases) and create reports.

3. What is a Data Lake? Explain the concept and describe how it differs from a data warehouse. When would a business prefer a data lake over a data warehouse?

- A data lake is a central storage environment used to store large amounts of data in original/raw form.
- Data warehouse stores processed and structured data, data lakes store raw data.
- Business would prefer data lakes if it has different forms of data to be processed later.

4. What is a Data Mart? How does it relate to a data warehouse and who typically uses it?

- Data Mart is a small, subject specific section of a data warehouse.
- Data Marts are used by specific departments because they provide users with easy access to relevant information.

5. Explain the difference between Structured, Semi-Structured, and Unstructured Data. Provide one example of each.

- Structured Data: data organised in fixed format, typically rows and columns (eg Excel spreadsheet).
- Semi Structured Data: not in rows and columns but tags or other information to help organise it (eg JSON file).
- Unstructured Data: Does not have defined structure (eg all eg video, emails, Audio files).



6. What is Data Modeling? Why is it important when designing a system that works with data? Describe at least two types of data models.

- Data Modelling is the process of designing how data will be organised, stored and related to other data in the system.
- Rational Model: Data organised into tables containing rows and columns. Relationship between tables are created using keys.
- Dimensional Model: Data organised using fact tables and dimension tables.

7. What is Data Mining? How is it used in business? Provide two real-world examples of data mining in action.

- Data Mining is the process of examining large amounts of data to identify useful patterns, trends, relationships and insights.
- Business use data mining to improve decision making, understand customers and predict future patterns.
- Banking: Bank analyse transactions to identify fraud.
- Retail: Stores analyse purchasing behaviours for promotions.

8. What is a Database Management System (DBMS)? Name and briefly describe four popular DBMS platforms (e.g., MySQL, PostgreSQL, Oracle, MongoDB) and explain a suitable use case for each.

- DBMS is a software/system that allows companies to create, store, manage, retrieve and update data in databases.
- MySQL: good for websites and business applications.
- Oracle Database: Suitable for organisation that require high performance, reliability, strong security.
- PostgreSQL: Suitable for business applications and analytical workloads.
- MongoDB: Suitable for data structures which change frequently.

9. What is ETL (Extract, Transform, Load)? Explain each step and describe why this process is important in data analytics.

- ETL is a process where data gets cleaned before it gets to data warehouse from data bases.
- Extract: Data is collected from different sources.
- Transform: Data is cleaned and changed useful and consistent format.
- Load: transformed data is loaded into target system such as data warehouse.



10. What are common data storage formats used in analytics (e.g., CSV, JSON, Parquet, Avro)? Briefly describe each and state when you would use one over another.

- **Comma-Separated Values (CSV)** Store data in rows and columns by commas.
- **JavaScript Object Notation (JSON)** Stores data using structure key-value format
- **Parquet**: A column-based file format designed for efficient storage and analytical processing
- **Avro**: A row-based data serialisation format that stores data together with its schema.



Section B: Artificial Intelligence & Machine Learning

Research Tip: Explore resources like Google's Machine Learning Crash Course, IBM's AI explanations, or OpenAI's blog to build a strong foundational understanding.

11. What is Artificial Intelligence (AI)? Define it and explain the difference between Narrow AI (Weak AI) and General AI (Strong AI). Provide one example of each.

- AI is the ability of computer systems to perform that normally require human intelligence, such as learning, reasoning, understanding language, decision making.
- **Narrow AI:** Narrow/Weak AI is designed to perform specific tasks, e.g., a spam filtering system can identify unwanted emails.
- **General AI:** Theoretical AI where computers learn patterns from data and use them to predict or make decisions.

12. What is Machine Learning (ML)? How does it differ from traditional programming? Describe the three main types of Machine Learning: Supervised, Unsupervised, and Reinforcement Learning.

- ML is a branch of AI where computers learn patterns from data and use those patterns to make predictions or decisions rather than being programmed for every situation.
- **Supervised Learning:** the model learns from labelled data.
- **Unsupervised Learning:** the model works with data that has no predefined labels.
- **Reinforced Learning:** An AI system learns by receiving rewards or penalties for its actions.

13. What is Deep Learning? How does it relate to Machine Learning and Artificial Intelligence? What are Neural Networks and how do they work?

- Deep Learning is a specific area of ML that uses artificial neural networks with multiple layers to learn complex patterns from large amounts of data.
- The relationship is represented as shown below:
Artificial Intelligence → Machine Learning → Deep Learning

14. What is Natural Language Processing (NLP)? Describe what it is and provide three real-world applications of NLP that you use in everyday life.

- NLP is a branch of AI that enables computers to understand, interpret, generate human language.
- **Voice assistants:** Siri and Google Assistant use NLP to understand spoken requests.
- **Chat bots:** use NLP to understand questions and provide responses.
- **Email spam filters:** analyse language and patterns to identify unwanted messages.



15. What is a Large Language Model (LLM)? How is it different from a standard Machine Learning model? Name two well-known LLMs and explain what they are used for.

- LLM is an AI model trained on very large amounts of text so that it can understand and generate human-like language
- ML is for specific tasks, LLM can perform many language related tasks, such as answering questions, summarising info, translating texts

16. What is Generative AI? How does it work at a high level? Name three tools that use Generative AI and describe what each one does.

- Gen AI is AI that can create new content based on patterns learned from existing data
- Gen AI is trained using large amounts of data, when a user provides an instruction or prompt
- ChatGPT: Generates conversational text and can assist with writing.

17. What is a Training Dataset? Why is the quality of training data so important when building an AI model? What could go wrong if the training data is biased or incorrect?

- A training dataset is a collection of examples used to teach a ML or AI model how to recognise patterns and provide useful outputs
- Important because AI model can only learn from data it receives
- Poor-quality training data can result in incorrect predictions, unfair decisions.

18. What is Computer Vision? How do machines 'see' and interpret images? Provide two real-world use cases of computer vision.

- Computer vision is a field of AI that enables computers to analyse and interpret images or videos
- Machines do this by processing the pixels in an image and using algorithms or neural networks to identify patterns

- Medical Language: AI can analyse X-rays, CT scans or other medical images
- Security: Computer vision can analyse CCTV footage and detect objects.



Section C: APIs, Tools & Developer Concepts

Research Tip: Search for 'What is an API for beginners' on YouTube or read documentation from platforms like Postman, RapidAPI, or Anthropic.

19. What is an API (Application Programming Interface)? Explain what it is using a simple real-world analogy. Why are APIs important in modern software and data systems?

- API is a way for different software applications to communicate with each other and exchange info.
- Simple real world analogy = waiter in ~~street~~ restaurant.
Customer tells waiter what they want. Waiter communicates with the kitchen and the brings food back to customer.
 - Customer: application making request
 - Kitchen: server/system
 - Waiter: API
 - Food: service.

20. What is an API Key? What is its purpose? How should it be stored and protected, and what could happen if it is exposed publicly?

- An API key is a unique piece of info used to identify and authenticate an application when it accesses an API.
- API keys should be stored securely using methods such as environment variables.
- If an API key is exposed publicly, another person could use it to access the service.

21. What is a REST API? What does REST stand for and what are the main HTTP methods (GET, POST, PUT, DELETE) used in REST APIs? Briefly describe each method.

- REST stands for Representational State Transfer.
- REST API is an API design that commonly uses HTTP to allow applications to communicate with resources over a network.
- GET: Retrieves info
- POST: Creates/submit new info
- PUT: updates/replaces existing info
- DELETE: Removes info

22. What is JSON (JavaScript Object Notation)? Why is it commonly used when working with APIs and data exchange? Write a simple example of a JSON object.

- JSON is a lightweight format used to represent and exchange structured data.
- JSON is commonly used because it is easy for humans to read and easy for computers to process.



23. What is Prompt Engineering? Why is it an important skill when working with AI tools like ChatGPT or Claude? Provide two examples of a poorly written prompt and an improved version of each.

- Prompt engineering is the process of designing clear and effective instructions for an AI system so that it produces a useful response.
- It is important because the quality of instructions can significantly affect the quality and usefulness of AI response.
- Poor Prompt: Tell me about risk.
- Improved Prompt: Explain risk in simple language.

24. What is a MAP function? Explain what it does in programming and data processing. Provide a simple example in either Python or JavaScript.

- A MAP function applies the same operation or transformation to every item in a collection, such as a list.
- Example:

```
numbers = [1, 2, 3, 4]
squared = list(map(lambda x: x * x, numbers))
print(squared)
```

Results: [1, 4, 9, 16]

25. What is Cloud Computing? Name the three main cloud service models (IaaS, PaaS, SaaS) and briefly explain each. Give one example of a product for each model.

- Cloud computing means using computing resources such as servers, storage, databases and software over the internet instead of having to own and maintain all the physical infrastructure yourself.
- IaaS - provides computing infrastructure such as virtual machines, storage, networking.
- PaaS - provides platforms where developers can build and run applications without managing infra.

26. What is a Version Control System? What is Git and why is it widely used by data analysts and developers? Briefly explain what a 'commit' and a 'branch' are.

- A Version Control System is software that records changes made to files over time.
- Git is one of the most widely used version control systems.
- Git is useful to data analysts and developers.



27. What is a Jupyter Notebook? What programming language is it most commonly used with? Why is it popular among Data Analysts and Data Scientists?

- A Jupyter Notebook is an interactive computing environment that allows users to combine programming code, explanations, charts, equations and results in one document.
- It is most commonly associated with Python.

28. What is Python? Why is it considered one of the most popular programming languages for Data Analytics and AI? Name four Python libraries commonly used in data work and briefly describe each.

- Python is a high level programming language known for being relatively easy to learn and widely used in data analytics, automation, AI and software development.
- Python is popular in data analytics because it has a large ecosystem of libraries for data manipulation, statistics, visualisation, Machine Learning.
- Common Python Libraries are:
 - > Pandas: used for working with and analysing structured data
 - > NumPy: used for ~~working with~~ numerical calculations
 - > Matplotlib: used to create charts and data visualisations
 - > Scikit-learn: used for ML tasks such as classification, regression and clustering



Section D: Data Ethics, Governance & Careers

Research Tip: Explore the World Economic Forum, GDPR documentation, and LinkedIn Learning for content on data ethics and career pathways in data.

29. What is Data Privacy? Why is it important? Research the GDPR (General Data Protection Regulation) and explain what it is and which countries it applies to.

- Data privacy is the responsible handling and protection of information about individuals. It is about collection, storing, using and sharing of ~~the~~ info appropriately.
- It is important because people should have control over how their personal info is used.
- GDPR is a European Union regulation that establishes rules for protecting personal data.

30. What is Data Ethics? Provide three examples of unethical use of data or AI, and explain what makes each example problematic.

- Data Ethics refers to the principles that guide the responsible collection, use, analysis and sharing of data.
- Examples:
 - Collecting personal information without proper consent
 - Using AI to discriminate.

31. What is Data Governance? Define it and explain the key components of a good data governance framework.

- Data Governance is the framework of policies, processes, responsibilities and controls used to ensure that an organisation is properly managed.
- Key components include:
 - Data ownership! Data Quality! Data Security! Data Privacy!

32. What is Bias in AI? Explain what it means for an AI model to be biased. Provide a real-world example of AI bias and describe the potential consequences.

- AI Bias happens when an AI system produces systematically unfair or inaccurate results for certain individuals or groups.
- AI Model is biased if it is trained to discriminate against certain individuals or groups. eg recent meet technology selecting English speaking people only.



33. What are the key differences between a Data Analyst, a Data Scientist, a Data Engineer, and an AI Engineer? Create a simple table comparing their roles, skills, and tools.

Role	Main Responsibility	Key Skills	Common Tools
Data Analyst	Analyses data and Create reports and insights	SQL, Excel, Stats, Visuals	Excel, SQL, Power BI, Tableau, Python
Data Scientist	Builds analytical and predictive models	Stats, ML, Python, Maths	Python, R, Jupyter
Data Engineer	Builds and maintains data pipelines and infra	Programming, Databases, ETC	SQL, Python, Spark, Cloud Platforms
AI Engineer	Builds and deploys AI and ML Solutions	ML, AI programming	Python, Cloud AI Platforms

34. What is Business Intelligence (BI)? Name two popular BI tools (e.g., Power BI, Tableau) and explain how they help businesses make data-driven decisions.

- BI is the processes and technologies used to transform data into information that helps organisations make better business decisions
- MS Power BI: Allows organisations to connect to data sources, analyse information and create interactive dashboards and reports

Section E: Personal Reflection

Answer the following reflection questions thoughtfully and honestly. There is no right or wrong answer — we want to know what you think!

35. Which concept from this assignment did you find most interesting or surprising? Explain why.

The concept I found most interesting is AI and ML. It's interesting how computers learn patterns from large data and can find patterns to make predictions or generate output.

36. Which concept did you find most difficult to understand? What resources did you use to help you understand it better?

The concept of the differences between data warehouse, data lake and data mart. Chat GPT and Copilot gave good examples of the difference.



37. How do you see Data and AI affecting your future career or everyday life? Be specific with examples.

Data and AI is having and will continue to have a significant impact in my risk management career. Data helps me make better decisions by identifying risk trends, risks and opportunities.

References

List all the sources you used to complete this assignment. Use the format below:

Author/Organisation. (Year). Title of article or webpage. Website Name. URL

1. Co-Pilot
2. Chat GPT
3. AI and R
- 4.
- 5.
- 6.
- 7.
- 8.

Good luck! We believe in you.

BrightLearn Data Analytics